

1. Let $J = \int_0^5 (t-1)e^{-t} dt$. If $u = t-1$ and $dv = e^{-t} dt$, then $du = dt$ and $v = -e^{-t}$ and integration by parts gives :

$$J = [-(t-1)e^{-t}]_0^5 - \int_0^5 -e^{-t} dt = [-te^{-t}]_0^5 = -5e^{-5}.$$

2. Let $J = \int \sin^3 x \sin 2x \cos^2 x dx = 2 \int \sin^4 x \cos^2 x \cos x dx$. The substitution $u = \sin x \Rightarrow du = \cos x dx$ gives : $J = 2 \int u^4(1-u^2) du = -\frac{2u^7}{7} + \frac{2u^5}{5} + C = -\frac{2\sin^7 x}{7} + \frac{2\sin^5 x}{5} + C$.

3. Let $I = \int_8^{16} \frac{dx}{\sqrt{x}-x}$. The substitution $u = \sqrt{x} \Rightarrow u^2 = x \Rightarrow 2udu = dx$ gives:

$$I = \int_{2\sqrt{2}}^4 \frac{2u du}{u-u^2} = 2 \int_{2\sqrt{2}}^4 \frac{du}{1-u} = -2 [\ln|1-u|]_{2\sqrt{2}}^4 = -2 [\ln|-3| - \ln|1-2\sqrt{2}|] \\ = 2 \ln \left(\frac{2\sqrt{2}-1}{3} \right).$$

4. Let $J = \int \frac{dx}{2+\cos x}$ by using the substitution $t = \tan(x/2)$ and $dx = \frac{2dt}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$ gives:

$$J = \int \frac{\frac{2dt}{1+t^2}}{2 + \frac{1-t^2}{1+t^2}} = \int \frac{2dt}{t^2+3} = \frac{2}{\sqrt{3}} \arctan \left(\frac{t}{\sqrt{3}} \right) + C = \frac{2}{\sqrt{3}} \arctan \left(\frac{\tan(x/2)}{\sqrt{3}} \right) + C$$

5. The limit is indeterminate of type 1^∞ since $\lim_{x \rightarrow 0} \operatorname{sech} x = 1$ and $\lim_{x \rightarrow 0} x^{-3/2} = \infty$. Let $y = (\operatorname{sech} x)^{x^{-3/2}}$. By continuity of the natural logarithmic function $\lim_{x \rightarrow 0} \ln y = \ln(\lim_{x \rightarrow 0} y)$. On the other hand,

$$\lim_{x \rightarrow 0} \ln y = \lim_{x \rightarrow 0} \frac{\ln(\operatorname{sech} x)}{x^{3/2}} =_{LH} \lim_{x \rightarrow 0} \frac{\frac{-\tanh x \cdot \operatorname{sech} x}{\operatorname{sech} x}}{\frac{3}{2}x^{1/2}} = \frac{2}{3} \lim_{x \rightarrow 0} \frac{-\sqrt{x} \cdot \tanh x}{\sqrt{x} \sqrt{x}} \\ = -\frac{2}{3} \left[\lim_{x \rightarrow 0} \sqrt{x} \cdot \lim_{x \rightarrow 0} \frac{\tanh x}{x} \right] = 0. \text{ We conclude that } \lim_{x \rightarrow 0} y = e^0 = 1.$$

6. We can transform the integrand into a function by first completing the square under the root sign then we obtain $\int \frac{dx}{\sqrt{(x-3)^2+4}}$. For trigonometric substitution, let $x-3 = 2 \tan \theta$ where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then $dx = 2 \sec^2 \theta d\theta$ and $\sqrt{(x-3)^2+4} = \sqrt{4 \tan^2 \theta + 4} = 2 \sec \theta$. Therefore

$$\int \frac{dx}{\sqrt{(x-3)^2+4}} = \int \frac{2 \sec^2 \theta}{2 \sec \theta} d\theta = \int \sec \theta d\theta = \ln(\tan \theta + \sec \theta) + C.$$

The right triangle gives,

$$\sec \theta = \frac{\sqrt{(x-3)^2+4}}{2}, \text{ so } \int \frac{dx}{\sqrt{(x-3)^2+4}} = \ln \left(\frac{x-3}{2} + \frac{\sqrt{(x-3)^2+4}}{2} \right) + C.$$

7. The first step is to divide then we obtain $\int \frac{x^4 - 2x}{x^4 - 1} dx = \int (1 + \frac{-2x+1}{x^4-1}) dx$. The second step is to factor the denominator then the form of the partial fraction decomposition is

$$\frac{-2x+1}{x^4-1} = \frac{A}{x-1} + \frac{B}{x+1} + \frac{Cx+D}{x^2+1}.$$

By solving we get $A = -\frac{1}{4}, B = -\frac{3}{4}, C = 1$ and $D = -\frac{1}{2}$, so

$$\begin{aligned} \int \frac{x^4 - 2x}{x^4 - 1} dx &= \int \left[x + \frac{-1/4}{x-1} + \frac{-3/4}{x+1} + \frac{x-1/2}{x^2+1} \right] dx \\ &= x - \frac{\ln|x-1|}{4} - \frac{3\ln|x+1|}{4} + \frac{\ln|x^2+1|}{2} - \frac{\arctan x}{2} + K. \end{aligned}$$

8. Let $J = \int_1^\infty x \arctan x dx$, which is improper so $J = \lim_{t \rightarrow \infty} \int_1^t x \arctan x dx$.

We integrate by parts with $u = \arctan x, dv = x dx$ then $du = \frac{1}{1+x^2} dx, v = \frac{x^2}{2}$:

$$\begin{aligned} \int_1^t x \arctan x dx &= \left[\frac{x^2}{2} \arctan x \right]_1^t - \frac{1}{2} \int_1^t \frac{x^2}{1+x^2} dx = \left[\frac{x^2}{2} \arctan x \right]_1^t - \frac{1}{2} \int_1^t \frac{x^2+1-1}{x^2+1} dx = \\ &= \left[\frac{x^2}{2} \arctan x - \frac{x}{2} + \frac{\arctan x}{2} \right]_1^t. \text{ Then} \end{aligned}$$

$$J = \lim_{t \rightarrow \infty} \left[\frac{x^2}{2} \arctan x - \frac{x}{2} + \frac{\arctan x}{2} \right]_1^t = \lim_{t \rightarrow \infty} \left[\frac{t}{2} (t \arctan t - 1) + \frac{\arctan t}{2} - \frac{\pi-2}{4} \right] = \infty.$$

So the given integral is divergent.